WATER TANK CONTROLLER

RTL-to-GDSII Implementation Final Report

Numerical results updated from the uploaded STA_All_Corners PDF

Project detail	Value
Top module	water_tank_controller
Technology	Sky130
Flow	OpenLane 1.0.2
Run	RUN_2026.10.10_02.11.36
Design type	Sequential / clocked RTL
Main interface	clk, rst_n, level_low, level_full, dry_run_fault, pump_en, alarm
STA source	Uploaded STA_All_Corners (3).pdf

Prepared using the actual numerical timing and power data visible in the uploaded STA report. DRC/LVS values are not claimed where the uploaded report does not establish them.

1. Abstract and Objectives

The Water Tank Controller is a synchronous Verilog RTL design implemented through an OpenLane RTL-to-GDSII flow using the Sky130 platform. It uses water-level and dry-run-fault status inputs to control pump-enable and alarm outputs. The uploaded STA document contains timing and power summaries for synthesis, physical-design stages, and extracted minimum, maximum and nominal RC corners.

Project objectives include implementing the controller in RTL, verifying sequential behaviour, synthesizing the design, completing physical implementation, reviewing multi-corner timing and power, and preserving the final layout and reports.

2. Design Overview

Signal	Direction	Purpose
clk	Input	Clock for sequential logic
rst_n	Input	Active-low reset
level_low	Input	Low water-level status
level_full	Input	Full tank status
dry_run_fault	Input	Dry-run fault status
pump_en	Output	Pump enable control
alarm	Output	Alarm output

The exact behaviour for simultaneous inputs should be described from the final RTL and testbench. This report does not infer an undocumented input priority.

3. RTL-to-GDSII Flow and Area

The uploaded reports identify the synthesis and physical-design flow stages, including CTS, placement, routing and extracted signoff STA.

Metric	Reported value	Source / note
Synthesis standard-cell count	14 cells	Synthesis AREA_0.stat.rpt
Synthesis chip area	157.6512 µm²	Synthesis AREA_0.stat.rpt
Technology	Sky130	STA report header
OpenLane version	1.0.2	STA report header
Final GDSII	Run artifact referenced	Run RUN_2026.10.10_02.11.36

The synthesis area and cell count are synthesis-stage values. They should not be presented as the final post-placement cell area unless the corresponding final metrics report confirms that.

4. Static Timing Analysis — Extracted RC Corners

Values below are copied from the signoff summary reports in the uploaded STA PDF: rcx_min_sta.summary.rpt, rcx_max_sta.summary.rpt and rcx_nom_sta.summary.rpt. Slack values are in nanoseconds.

RC corner	TNS	WNS	Worst setup slack	Worst hold slack
Minimum RC (rcx_min)	0.00	0.00	10.07 ns	0.13 ns
Maximum RC (rcx_max)	0.00	0.00	10.03 ns	0.14 ns
Nominal RC (rcx_nom)	0.00	0.00	10.05 ns	0.14 ns

Interpretation

TNS = 0.00 and WNS = 0.00 are reported for all three extracted corners. The worst setup slack is positive at each corner (10.03–10.07 ns), and the worst hold slack is also positive (0.13–0.14 ns) in these summaries. These values indicate positive reported slack for the paths covered by these reports; they do not independently prove that every design constraint is complete or that every signoff check passed.

5. Power Analysis — Extracted RC Corners

Power values are from the corresponding extracted signoff power reports. Values are shown in watts and rounded as printed in the source report.

RC corner	Internal	Switching	Leakage	Total
Minimum RC	3.91e-05 W	8.43e-06 W	1.52e-08 W	4.75e-05 W
Maximum RC	3.91e-05 W	8.92e-06 W	1.52e-08 W	4.81e-05 W
Nominal RC	3.91e-05 W	8.67e-06 W	1.52e-08 W	4.78e-05 W

The reported total estimated power is approximately 47.5–48.1 µW across the three extracted RC corners. These are tool estimates under the activity and constraints used by the run, not measurements from fabricated hardware.

6. Timing Checks and Parasitic Annotation

Check	Observed result in uploaded report
Maximum slew violations	0
Maximum fanout violations	0
Maximum capacitance violations	0
Unannotated drivers	28 in CTS checks report
Partially unannotated drivers	0 in CTS checks report

The CTS check report lists 28 unannotated drivers, including clock, input, output and internal nets. This should be disclosed when discussing the scope of timing/power annotation.

7. DRC, LVS and Signoff

The uploaded STA PDF contains timing and power reports and references signoff artifacts, but the extracted numerical evidence reviewed for this report does not establish final DRC violation count or LVS error count. Therefore, no zero-violation claim is made here. Insert the exact DRC and LVS results from the matching Water Tank Controller run before final submission.

8. Simulation and Verification

RTL verification should exercise reset, low-level response, full-tank response, dry-run fault handling and combinations of status inputs. Simulation pass counts and waveform evidence were not numerically established by the STA PDF, so they are not invented in this report.

9. Recommended Repository Contents

Path	Contents
rtl/	Water Tank Controller Verilog RTL
simulation/	Testbench, simulation output and waveform
openlane/water_tank_controller/	OpenLane configuration and run-specific files
results/sta/	STA reports grouped by synthesis and extracted RC corner
results/final/	Curated final GDSII/LEF deliverables
results/signoff/	DRC/LVS and other signoff reports
screenshots/	RTL, simulation, waveform and layout screenshots
docs/	This final report and results_summary.txt

10. Results Summary and Conclusion

Metric	Value
Technology / flow	Sky130 / OpenLane 1.0.2
Synthesis cell count	14
Synthesis chip area	157.6512 µm²
Min RC TNS / WNS	0.00 / 0.00
Min RC setup / hold slack	10.07 ns / 0.13 ns
Max RC setup / hold slack	10.03 ns / 0.14 ns
Nominal RC setup / hold slack	10.05 ns / 0.14 ns
Extracted total power range	4.75e-05 to 4.81e-05 W
Slew / fanout / capacitance violations	0 / 0 / 0 in CTS checks report
DRC / LVS	Not established by reviewed numerical evidence

Conclusion

The uploaded Water Tank Controller STA report supports numerical reporting of positive extracted setup and hold slack, zero TNS/WNS, power estimates across minimum/maximum/nominal RC corners, synthesis area and cell count, and zero slew/fanout/capacitance violations in the CTS checks report. DRC/LVS status and simulation pass results remain unclaimed until their matching reports are checked.